

Case - Infeasible Model

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:53

Summary flowchart branch: row 0 has no negative values but an artificial variable remains in the basis at a positive level -> no feasible solution exists.

Model

Maximize $Z = 3X_1 + 2X_2$
subject to
[C1] $2X_1 + X_2 \leq 2$
[C2] $3X_1 + 4X_2 \geq 12$
 $X_1, X_2 \geq 0$

RESULT

INFEASIBLE

iterations: 1 · rule: dantzig

Findings

- Infeasible: row 0 has no negative values but artificial variable(s) X_5 remain in the basis at a positive level.

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) $\rightarrow$ $X3$ is the starting basic variable of this row.
2. C2 (Type II, $\geq$): subtract surplus $X4 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis $\rightarrow$ an artificial variable is needed.
3. C2: add artificial $X5 \geq 0$ with objective coefficient $-M$ (huge penalty) $\rightarrow$ it provides an immediately recognisable starting basic variable.

Maximize $Z = 3X1 + 2X2 - MX5$

subject to

$$2X1 + X2 + X3 = 2$$

$$3X1 + 4X2 - X4 + X5 = 12$$

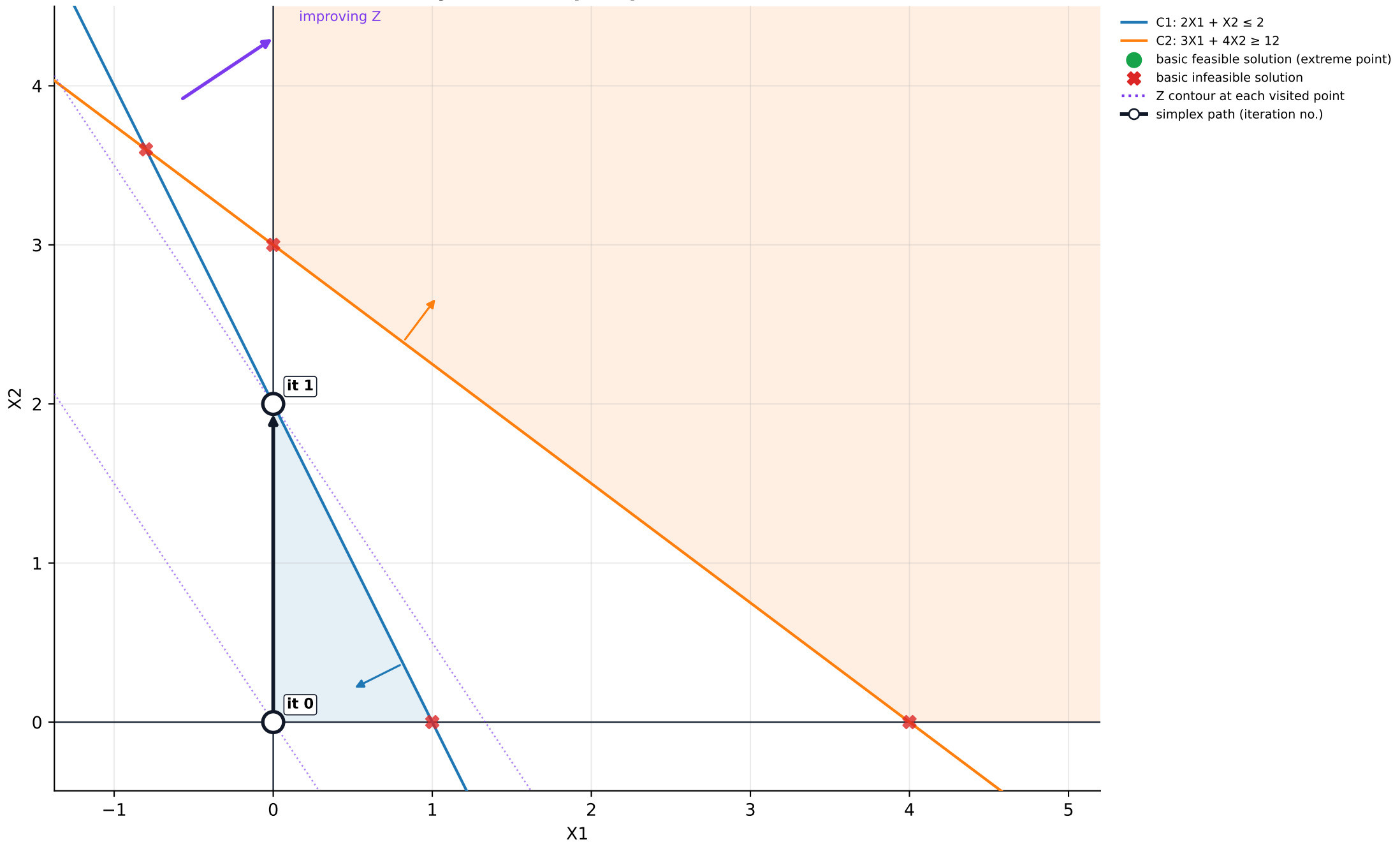
$$X1, X2, X3, X4, X5 \geq 0$$

Standard form - coefficient matrix

	X1	X2	X3	X4	X5		RHS
max Z	3	2	0	0	-M		
C1	2	1	1	0	0	=	2
C2	3	4	0	-1	1	=	12

2. Geometric view: basic solutions, extreme points and the simplex path

Geometry of the simplex path - infeasible



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	Optimal?
X1, X2	-0.8	3.6	no	4.8	0	0	
X1, X3	4	0	no	12	-6	0	
X1, X4	1	0	no	3	0	-9	
X2, X3	0	3	no	6	-1	0	
X2, X4	0	2	no	4	0	-4	
X3, X4	0	0	no	0	2	-12	

Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS
Z	0	1	-3	-2	0	0	M	0
X3	1	0	2	1	1	0	0	2
X5	2	0	3	4	0	-1	1	12

Initial tableau. Basic variables: $X3 = 2$, $X5 = 12$. Non-basic (set to 0): $X1$, $X2$, $X4$. The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

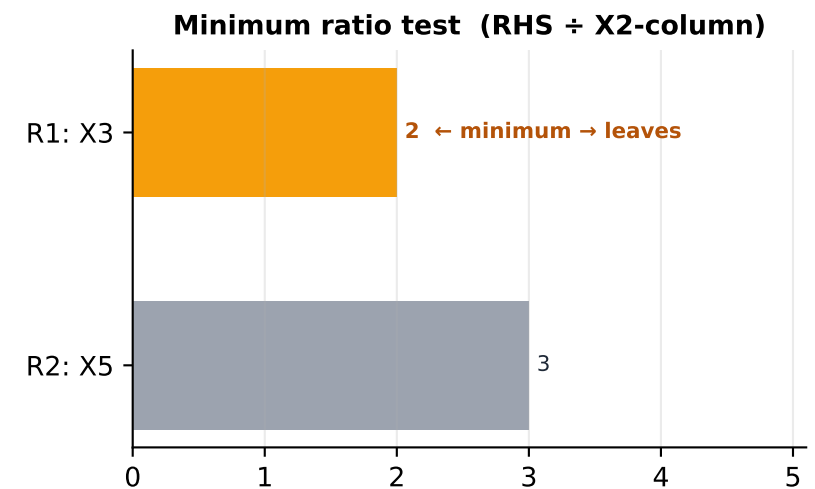
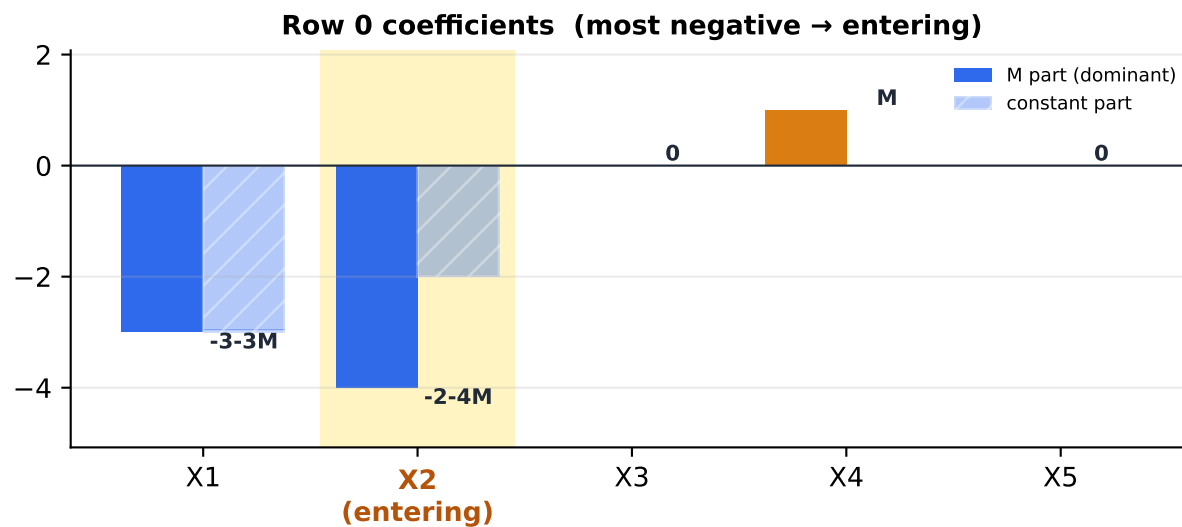
Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS
Z	0	1	-3-3M	-2-4M	0	M	0	-12M
X3	1	0	2	1	1	0	0	2
X5	2	0	3	4	0	-1	1	12

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting M × their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract M × R2 from R0 (make the coefficient of basic X5 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS	Ratio
Z	0	1	-3-3M	-2-4M	0	M	0	-12M	
X3	1	0	2	1	1	0	0	2	2
X5	2	0	3	4	0	-1	1	12	3



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 ($-2-4M$).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: 2, R2: 3.
- Leaving basic variable: X3 (row 1, smallest ratio). Pivot number = 1.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Pivot number is already 1 - scaling not needed (normally divide the pivot row by the pivot number)

Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS	Ratio
Z	0	1	-3-3M	-2-4M	0	M	0	-12M	
X3	1	0	2	1	1	0	0	2	2
X5	2	0	3	4	0	-1	1	12	3

(b) X2 replaced X3 in the basis. $Z = 4-4M$

Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS
Z	0	1	1+5M	0	2+4M	M	0	4-4M
X2	1	0	2	1	1	0	0	2
X5	2	0	-5	0	-4	-1	1	4

Row operations:

- Add $(2+4M) \times R1$ to $R0$
- Subtract $4 \times R1$ from $R2$

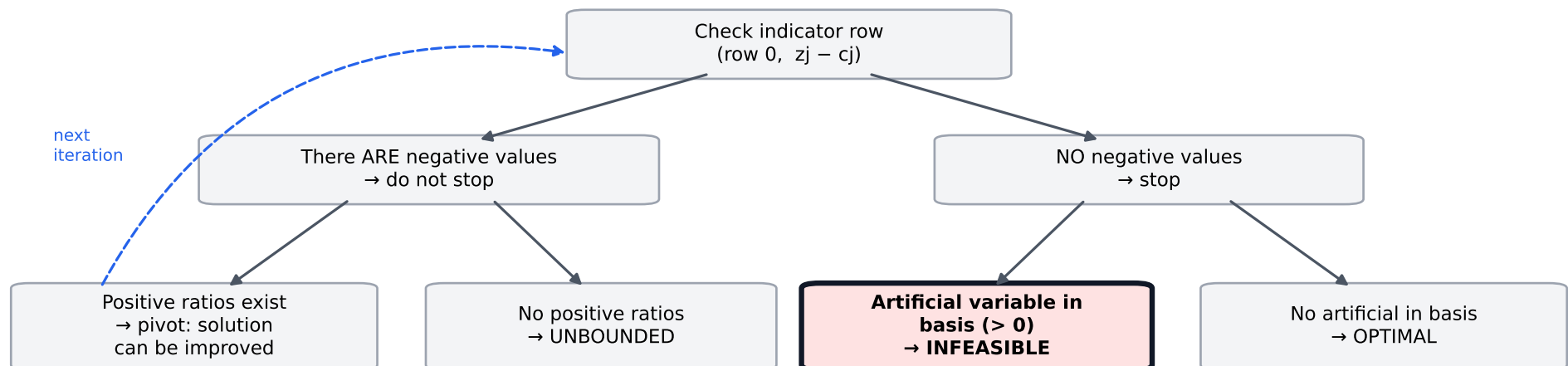
Final tableau - INFEASIBLE

Basic	Eq.	Z	X1	X2	X3	X4	X5	RHS
Z	0	1	$1+5M$	0	$2+4M$	M	0	$4-4M$
X2	1	0	2	1	1	0	0	2
X5	2	0	-5	0	-4	-1	1	4

Final check.

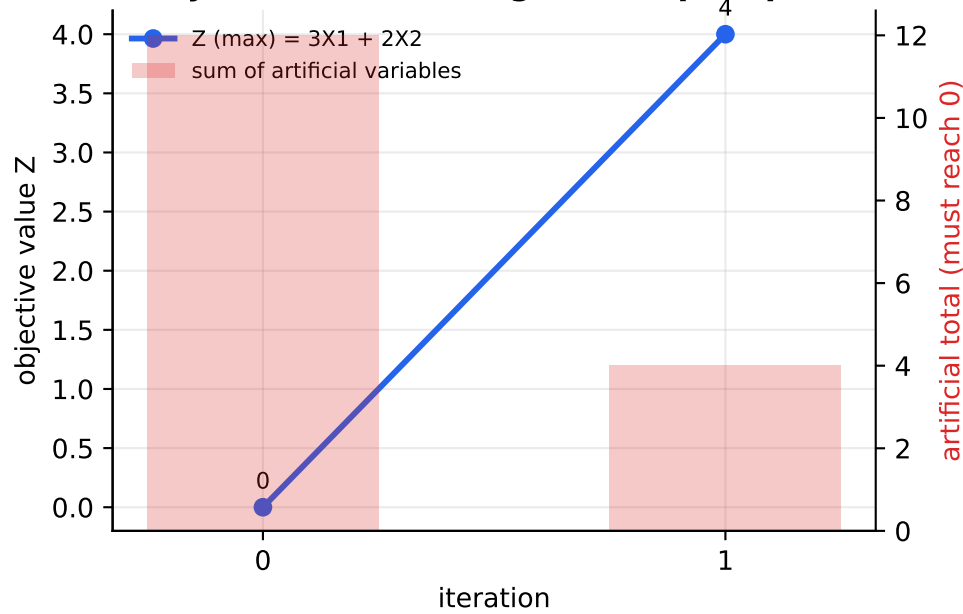
- Infeasible: row 0 has no negative values but artificial variable(s) X5 remain in the basis at a positive level.

Simplex decision diagram - outcome: INFEASIBLE

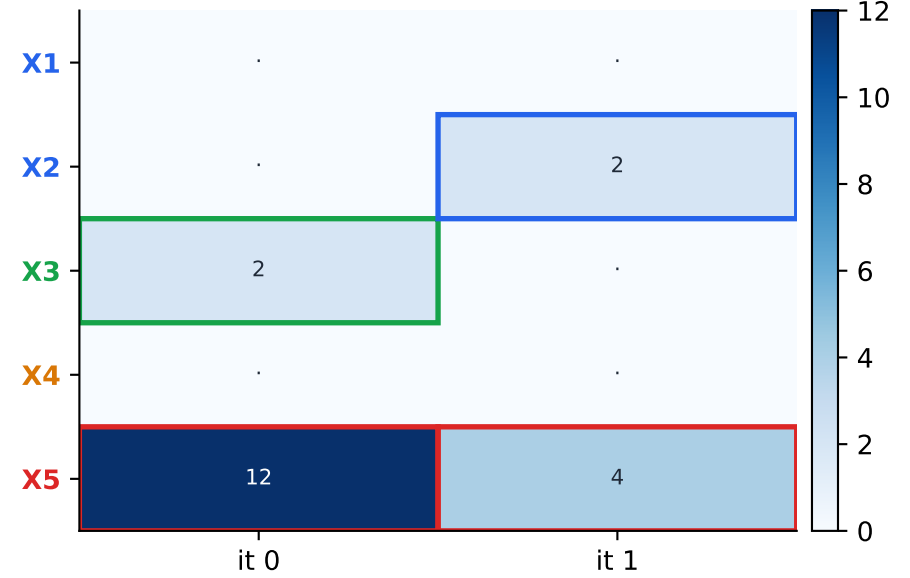


Case - Infeasible Model - INFEASIBLE

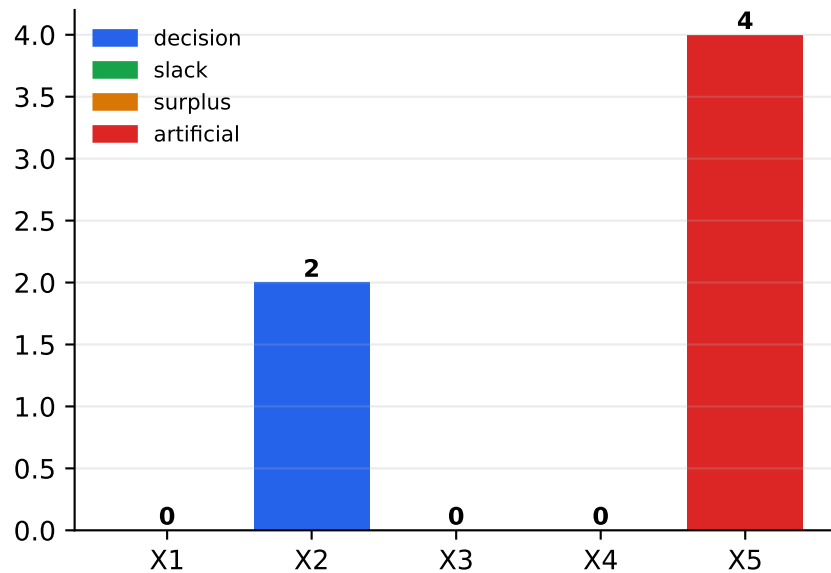
Objective value along the simplex path



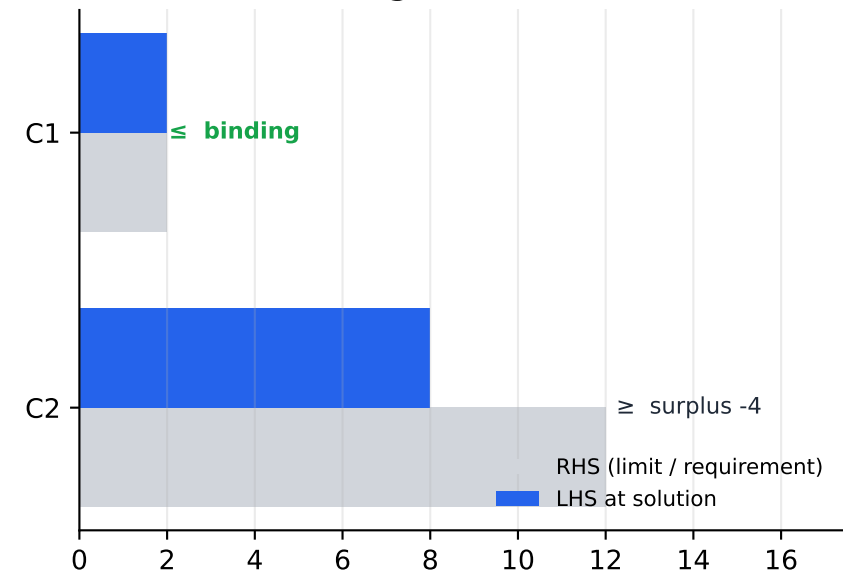
Basis evolution (outlined = basic variable)



Final values of all variables (infeasible)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	infeasible	infeasible

✓ **Results agree**